

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2_AT2 — VISUALIZATION DEVELOPMENT EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Development (BL4)	Assessment:	Assessment Tool 2 – Visualization Development
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO2 Statement:	<i>Analyze and synthesize multiple data views into a coherent, annotated dashboard that communicates a clear narrative insight. (BL4)</i>		
Student Name:	RAGUL.M	Reg. No.:	192524066
Section / Batch:	SLOT A	Date:	5/08/2026

Code of Conduct

I RAGUL.M[192524066] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: RAGUL.M

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must include at least two coordinated charts sharing a common axis or theme.
- Include at least one annotation calling out a key data point in your dashboard.
- Write a one-sentence narrative takeaway summarizing the insight your dashboard reveals.

Annexure A – VISUALIZATION DEVELOPMENT

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT2 — Visualization Development Exercise

CO2 · BT4: Analyze ·

This rubric is used to assess student responses to the CO2-AT2 Visualization Development Exercise, in which students build a small multi-chart dashboard from given data, add at least one annotation, and write a narrative takeaway.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of which chart types to combine, how they relate to each other, and how annotation and narrative should be used to communicate insight.	Good understanding of chart combination and annotation/narrative technique, with only minor gaps.	Basic understanding of dashboard development, but with some misconceptions about which charts to combine or how to annotate them.	Poor understanding of core development concepts; combines charts or annotations that do not suit the data or the story being told.
Explanation	25%	The annotation and narrative takeaway are precise, well-justified, and clearly tied to a specific, correctly identified data point or pattern.	The annotation and narrative are clear and reasonably well-justified, with adequate connection to the data.	The annotation or narrative is present but vague, generic, or weakly connected to the actual data shown.	Annotation or narrative is missing, incorrect, or unconnected to the data in the dashboard.
Application	20%	Effectively applies multi-chart coordination, annotation, and storytelling techniques to	Applies development techniques to produce a workable dashboard, with only minor errors or missed	Limited application; the dashboard partially addresses the scenario but charts feel disconnected or the story is unclear.	Fails to apply appropriate development techniques; the charts, annotation, and narrative do not

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		produce a complete, cohesive dashboard for the scenario.	refinements.		work together as a dashboard.
Organization	15%	Charts are clearly coordinated (aligned axes or shared theme), laid out logically, with the annotation and narrative placed for maximum clarity.	Charts are organized and mostly coordinated, with only minor issues in alignment or layout.	Basic structure; charts are present but poorly aligned or the annotation/narrative feels disconnected from the layout.	Poorly organized; charts are not coordinated, and the annotation or narrative is missing or misplaced.
Accuracy	10%	All chart values, scales, and the annotated data point are completely accurate and correctly reflect the given data.	Mostly accurate, with only minor omissions or a small error in one chart or the annotation.	Partially correct; some plotted values or the annotated point are missing or incorrect.	Largely inaccurate; major errors in plotted values, scales, or the annotated data point.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

21. Nonprofit Fundraising Dashboard

Problem Statement:

Nonprofit organizations collect donations through multiple channels, but it can be difficult to identify which channel provides the best results. A visual dashboard is required to compare donation performance and donor retention.

Methodology:

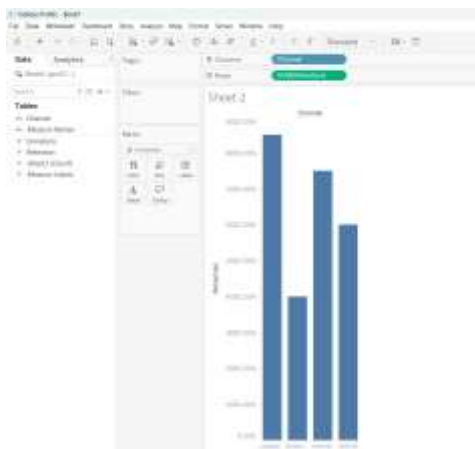
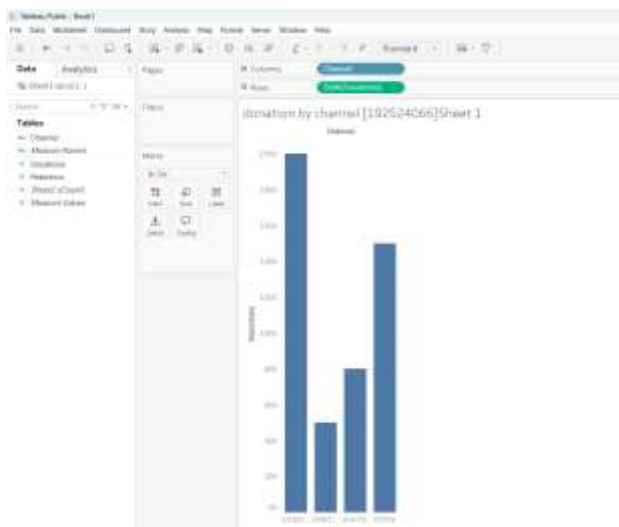
The fundraising dataset is imported into Tableau. The data is categorized according to fundraising channels. Donation values and retention rates are represented using separate visualizations. The visualizations are combined into an interactive dashboard.

Expected Outcome:

The dashboard clearly identifies the highest-performing fundraising channel and provides a quick comparison of all channels.

Business Insight:

Corporate fundraising provides strong financial performance along with high donor retention. Improving weaker channels can further increase overall fundraising efficiency.



22. Real Estate Market Dashboard

Problem Statement:

Real estate data contains different property prices and selling durations. Without visualization, it is difficult to compare neighborhoods and understand market demand.

Methodology:

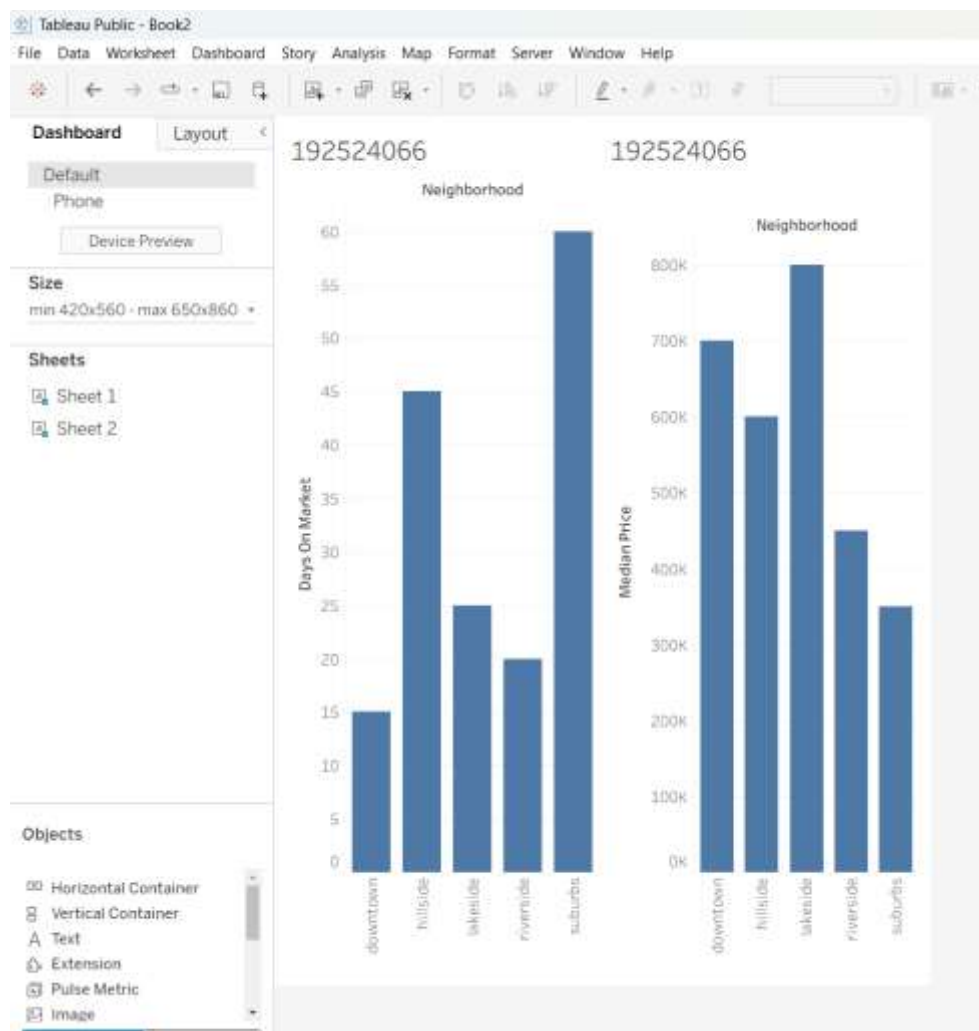
The neighborhood dataset is imported into Tableau. Median prices and days on market are analyzed separately using bar charts. The charts are combined into a dashboard for comparison.

Expected Outcome:

The dashboard helps users identify expensive areas, affordable areas, and locations where properties are sold more quickly.

Business Insight:

A neighborhood with a high property price and low days on market can indicate strong market demand. This information can support property buying and selling decisions.



23. Fitness App Cohort Dashboard

Problem Statement:

Fitness applications may experience user drop-off after registration. It is important to determine which user cohorts have poor long-term retention.

Methodology:

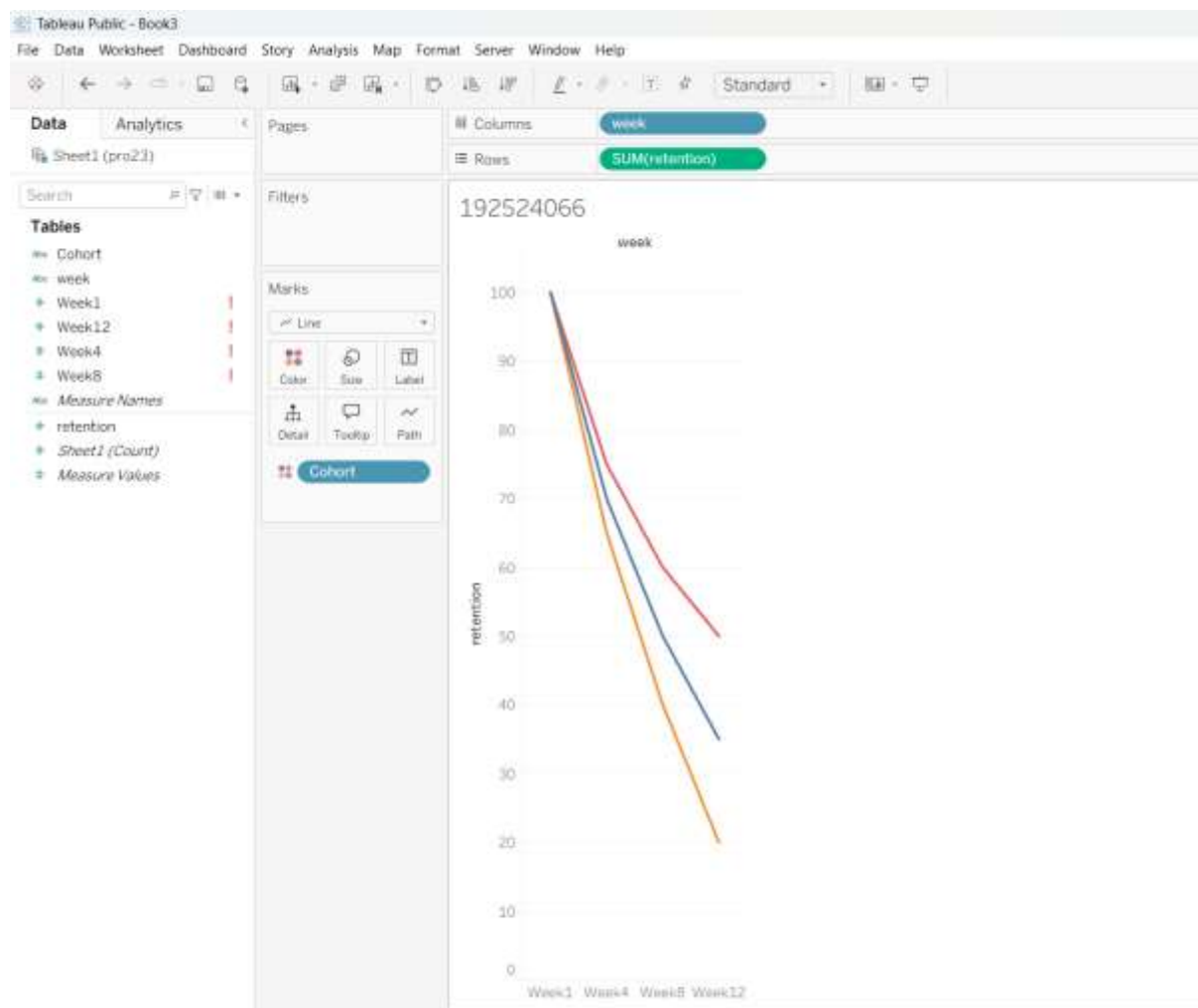
User retention data is organized by cohort and week. The weekly values are transformed using Tableau's pivot functionality. A line chart is created to compare retention trends among cohorts.

Expected Outcome:

The dashboard makes it easier to identify retention patterns and the point where significant user drop-off occurs.

Business Insight:

The March cohort maintains better long-term retention, while February shows a greater decline. The company can use this information to improve onboarding and engagement.



24. Weather Storm Tracking Dashboard

Problem Statement:

Storm conditions can change rapidly. Monitoring only individual measurements makes it difficult to understand the overall development of a storm.

Methodology:

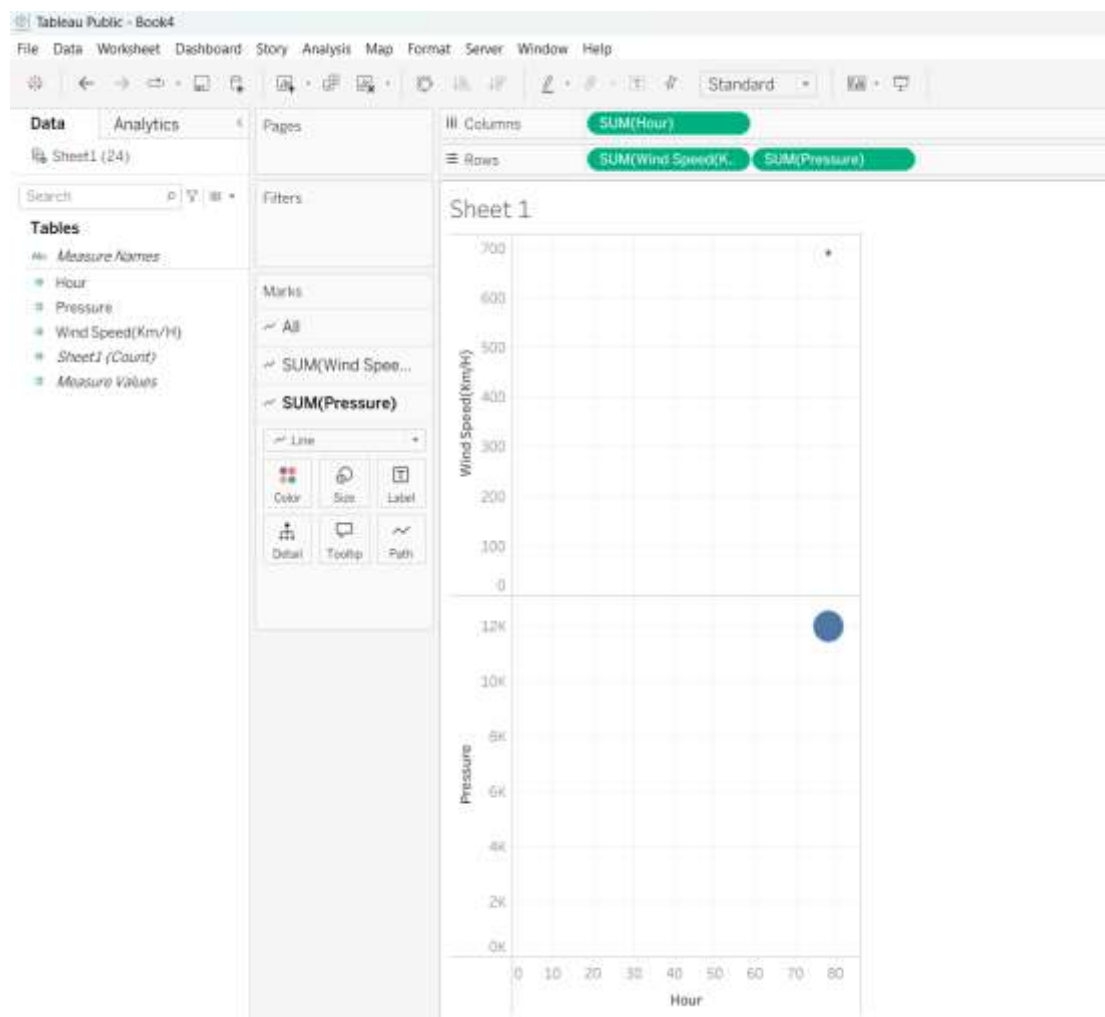
Hourly wind speed and pressure data are imported into Tableau. Time is placed on the X-axis, while the two weather measurements are visualized using line charts and a dual-axis chart.

Expected Outcome:

The dashboard identifies the period of maximum storm intensity and shows whether the storm is strengthening or weakening.

Business Insight:

The strongest conditions occur around Hour 6, when wind speed is highest and atmospheric pressure is lowest. The later reduction in wind speed indicates weakening conditions.



25. Company-Wide KPI Dashboard

Problem Statement:

Companies generate large amounts of performance data. Managers need a simple way to monitor revenue, customer growth, and employee growth together.

Methodology:

The four-year KPI dataset is imported into Tableau. Separate line charts are created for revenue, customers, and employees. These visualizations are combined into a single dashboard.

Expected Outcome:

The dashboard provides a clear overview of business growth and allows management to compare KPI trends across different years.

Business Insight:

The company shows consistent growth in all three KPIs. Customer growth is particularly significant, while revenue also increases substantially.

